

Air Quality Forecasting

2 Outputs Visualizations

Task 1: Air Quality Forecasting — Outputs & Visualizations

Executed using the supplied UCI Air Quality dataset.

```
import pandas as pd, numpy as np, matplotlib.pyplot as plt
DATA_PATH="/mnt/data/AirQualityUCI_submission.csv"
raw=pd.read_csv(DATA_PATH,sep=None,engine="python")
print("Rows:", raw.shape[0], "Columns:", raw.shape[1])
print("Date range:", raw["Date"].min(), "to", raw["Date"].max())
print("\nMissing-value code (-200):")
print((raw==-200).sum().sort_values(ascending=False).to_string())
```

Descriptive statistics

```
numeric=raw.select_dtypes(include=np.number).columns
display(raw[numeric].replace(-200,np.nan).describe().T.round(3))

df=raw.copy()
df["Datetime"]=pd.to_datetime(df["Date"].astype(str)+" "+df["Time"].astype(str))
df=df.sort_values("Datetime").reset_index(drop=True)
num=df.select_dtypes(include=np.number).columns
df[num]=df[num].replace(-200,np.nan).ffill()
df["hour"]=df["Datetime"].dt.hour
df["day_of_week"]=df["Datetime"].dt.dayofweek
df["month"]=df["Datetime"].dt.month
print("Remaining missing values after causal forward-fill:")
print(df.isna().sum().sort_values(ascending=False).head(10).to_string())
```

Visualization 1 — CO over time

```
fig,ax=plt.subplots(figsize=(13,5))
ax.plot(df["Datetime"],df["CO(GT)"],linewidth=0.8)
ax.set_title("Ground-Truth CO Concentration Over Time")
ax.set_xlabel("Date and time"); ax.set_ylabel("CO(GT)"); ax.grid(alpha=0.25)
plt.tight_layout(); plt.show()
```

Visualization 2 — CO distribution

```
fig,ax=plt.subplots(figsize=(8,5))
ax.hist(df["CO(GT)"].dropna(),bins=40)
ax.set_title("Distribution of Ground-Truth CO"); ax.set_xlabel("CO(GT)"); ax.set_ylabel("Frequency")
plt.tight_layout(); plt.show()
```

Visualization 3 — Average CO by hour

```
hourly=df.groupby("hour")["CO(GT)"].mean()
fig,ax=plt.subplots(figsize=(9,4))
ax.plot(hourly.index,hourly.values,marker="o")
ax.set_title("Average CO(GT) by Hour of Day"); ax.set_xlabel("Hour"); ax.set_ylabel("Mean CO")
ax.set_xticks(range(24)); ax.grid(alpha=0.25); plt.tight_layout(); plt.show()
print("Highest average hour:",int(hourly.idxmax()),"Mean:",round(hourly.max(),3))
print("Lowest average hour:",int(hourly.idxmin()),"Mean:",round(hourly.min(),3))
```

Visualization 4 — Average CO by day of week

```
dow=df.groupby("day_of_week")["CO(GT)"].mean()
labels=["Mon","Tue","Wed","Thu","Fri","Sat","Sun"]
fig,ax=plt.subplots(figsize=(8,4))
ax.bar(range(7),dow.reindex(range(7)).values)
ax.set_title("Average CO(GT) by Day of Week"); ax.set_xlabel("Day"); ax.set_ylabel("Mean CO")
ax.set_xticks(range(7),labels); plt.tight_layout(); plt.show()
```

Visualization 5 — Correlation matrix

```
cols=["CO(GT)","NOx(GT)","NO2(GT)","C6H6(GT)","T","RH","AH"]
corr=df[cols].corr()
print(corr["CO(GT)"].sort_values(ascending=False).round(3).to_string())
fig,ax=plt.subplots(figsize=(9,7))
im=ax.imshow(corr,aspect="auto")
ax.set_xticks(range(len(cols)),cols,rotation=70,ha="right")
ax.set_yticks(range(len(cols)),cols); ax.set_title("Correlation Matrix")
plt.colorbar(im,ax=ax); plt.tight_layout(); plt.show()
```

Visualization 6 — Monthly average CO

```
monthly=df.groupby("month")["CO(GT)"].mean()
fig,ax=plt.subplots(figsize=(8,4))
ax.plot(monthly.index,monthly.values,marker="o")
ax.set_title("Average CO(GT) by Month Number"); ax.set_xlabel("Month"); ax.set_ylabel("Mean CO")
ax.grid(alpha=0.25); plt.tight_layout(); plt.show()
```

Feature engineering output

```
df["target_next_CO"]=df["CO(GT)"].shift(-1)
for c in ["CO(GT)", "NOx(GT)", "NO2(GT)", "C6H6(GT)", "T", "RH", "AH"]:
    df[f"{c}_lag1"]=df[c].shift(1)
    df[f"{c}_lag3"]=df[c].shift(3)
    df[f"{c}_lag24"]=df[c].shift(24)
for c in ["CO(GT)", "NOx(GT)", "NO2(GT)", "C6H6(GT)"]:
    df[f"{c}_roll3"]=df[c].shift(1).rolling(3).mean()
    df[f"{c}_roll24"]=df[c].shift(1).rolling(24).mean()
display(df[["Datetime", "CO(GT)", "target_next_CO", "CO(GT)_lag1", "CO(GT)_lag3", "CO(GT)_lag24", "CO(GT)_roll3", "CO(GT)_roll24", "hour", "day"]])
```